Synchronous Machine Parameters

X_d direct axis reactance
X_q quadrature axis reactance
X'_d direct axis transient reactance
X'_q quadrature axis transient reactance
X''_d direct axis subtransient reactance
X''_q quadrature axis subtransient reactance
X_2 negative sequence reactance
X_0 zero sequence reactance

r_{sdc} stator dc resistance
r_{sac} stator ac resistance
$r_{fr'}$ field resistance referred to the stator
r_2 negative sequence resistance

T'_{d0} direct axis open-circuit transient time-constant
T'_d direct axis short-circuit transient time-constant
T''_d direct axis short-circuit subtransient time-constant
T_a armature short-circuit (d.c.) time-constant

Hydro Generator (salient pole rotor)

$$X_{md} = \omega_B \frac{kN_s}{N_D} kM_D \qquad X_{mq} = \omega_B \frac{kN_s}{N_D} kM_Q$$

$$X_d = \omega_B L_d = X_{md} + X_{lds}$$

$$X_q = \omega_B L_q = X_{mq} + X_{lqs}$$

$$X'_d = \frac{X_{md} X_{lF'}}{X_{md} + X_{lF'}} + X_{lds}$$

$$X'_q = X_{mq} + X_{lds}$$

$$X''_d = \frac{X_{md} X_{lD'} X_{lF'}}{X_{md} X_{lF'} + X_{lF'} X_{lD'} + X_{lD'} X_{md}} + X_{lds}$$

$$X''_q = \frac{X_{mq} X_{lQ'}}{X_{mq} + X_{lQ'}} + X_{lqs}$$

$$X_2 = \frac{X''_d + X''_q}{2}$$

$$X_0 = \frac{X_{ld} + X_{lq}}{2}$$

$$T'_{d0} = \frac{1}{R_{F'}} \left\{ \frac{X_{md} + X_{lF'}}{\omega_b} \right\}$$

$$T'_d = \frac{1}{\omega_b R_{F'}} \left\{ \frac{X'_d X_{lds}}{X_{md} + X_{lds}} + X_{lF'} \right\}$$

$$T'_d = \frac{X'_d}{X_d} T'_{do}$$

$$T''_d = \frac{X''_d}{X_d} T''_{do}$$

$$T''_q = \frac{X''_q}{X_q} T''_{qo}$$

$$T_a = \frac{1}{r_{sac}\omega_b} \left\{ \frac{2X''_d X''_q}{X''_d + X''_q} \right\}$$

Machine Constant	Turbo Generator	Hydro Generator	Synchronous Condensor	Synchronous Motor
X_d	1.1	1.15	1.80	1.20
X_q	1.08	0.75	1.15	0.90
X'_d	0.23	0.37	0.40	0.35
X'_q	0.23	0.75	1.15	0.90
X''_d	0.12	0.24	0.25	0.30
X''_q	0.15	0.34	0.30	0.40
X_2	0.13	0.29	0.27	0.35
X_0	0.05	0.11	0.09	0.16
r_{sdc}	0.003	0.012	0.008	0.01
r_{sac}	0.005	0.012	0.008	0.01
r_2	0.035	0.100	0.05	0.06
T'_{d0}	5.6	5.6	9.0	6.0
T'_d	1.1	1.8	2.0	1.4
T''_d	0.035	0.035	0.035	0.035
T_a	0.16	0.15	0.17	0.15